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Research Article

INTERNATIONAL JOURNAL OF ADVANCED ROBOTIC SYSTEMS

International Journal of Advanced Robotic Systems
March-April 2018: 1–11
© The Author(s) 2018
DOI: 10.1177/1729881418770545
journals.sagepub.com/home/arx
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Vehicle detection framework for challenging lighting driving environment based on feature fusion method using adaptive neuro-fuzzy inference system

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Abstract

This article proposes a new preceding vehicle detection framework for challenging lighting environments using a novel feature fusion technique based on an adaptive neuro-fuzzy inference system. A combination of two feature descriptors, the histogram of oriented gradients and local binary patterns, is adopted to improve the vehicle detection accuracy of the proposed framework, and the performance of the combination in image transformations is evaluated. Furthermore, we tested the detection performance of the proposed framework in three challenging driving conditions and filmed the test image sequences for each categorized environment of the experiments. The experimental results demonstrate that the proposed framework outperforms the conventional framework under specific driving environments with harsh lighting conditions.

Keywords

Visual object detection, vehicle detection, binary descriptor, feature fusion, adaptive neuro-fuzzy inference system

Date received: 24 November 2017; accepted: 20 March 2018

Topic: Vision Systems
Topic Editor: Antonio Fernandez-Caballero
Associate Editor: Grazia Cicirelli

Introduction

Over the last few decades, visual object detection based on monocular vision has been a subject of controversy in diverse areas, such as surveillance,[1] robotics,[2] medical devices,[3] and unmanned aerial vehicles.[4] Among a variety of areas, advanced driver assistant systems (ADASs) are one of the most widely studied subjects.[5–9] It is clear that visual information is essential for enabling such systems to work correctly. ADAS exists in various driving environments, so the purpose of visual object detection in ADAS is to recognize vehicle objects robustly in this circumstance. To solve the visual object detection problem, monocular cameras play an important role in obtaining visual information and have been extensively used for visual object tracking systems.[10,11] In visual object detection, several key issues exist. Vehicles on the road are typically in motion effects of ego and relative motion. There is variability in the size, shape, and color of vehicles encountered on the road. The on-road environment also features variations in illumination, background, and scene complexity.[12]

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In this article, we focus on robustly detecting vehicles in front of the driver using monocular camera under harsh lightning condition.

Prior to the explanation of the special vehicle detection conditions applied to our framework, we briefly introduce the trends of vehicle detection methods. Vehicles on the road are of various colors, shapes, and sizes.[12] The driving environment also has diverse background, illumination, and scene complexity. Most recent vehicle detection frameworks can be divided into two major categories: the appearance-based method and the model-based method. The appearance-based method is generally composed of two parts: constructing an object model by gathering a set of reference images and classifying images by comparing the parts of input images with those of reference images. Histogram matching,[13] edge matching,[14] gradient matching,[15] and divide-and-conquer searching are classified as the appearance-based method. To detect a vehicle that is located in a blind spot, an algorithm using a histogram of oriented gradients (HOG) feature and a deformable part–based model has been mainly proposed.[16] Also, a vehicle detection algorithm using a Haar-like feature and adaptive boosting (AdaBoost) is suggested to detect a vehicle.[17] Furthermore, the model-based method has been actively studied with various feature extraction and description methods and is applied to various vision-based systems. This approach contains two phases: a model construction (or training) phase that uses learning methods and a detection phase for classifying input images. There are, in general, strengths to the object detection method based on the object model. First, learning is easy since the learning process is performed automatically by well-designed learning methods, such as artificial neural networks,[18] support vector machines,[19] and AdaBoost.[20] Second, it shows general performance in contrast to the appearance-based model, since objects are depicted as a set of features that are robust to geometric primitives (e.g. lines and circles). Optical flow and Kalman filter algorithms have been proposed for tracking an approaching vehicle from the blind spot.[21] To detect vehicles in driving images, optical flow and hidden Markov model methods have been suggested.[22]

In the object detection framework based on the object model, the importance of feature type selection is indisputable. Numerous features have been adopted to increase the accuracy of the object detection system. Diverse image features, such as the HOG,[23] Haar features, and local binary pattern (LBP)[24] are adapted by object detection systems.[25,26] The different approaches to image features are as follows. Dollar et al. suggested a method of approximating multi-resolution image features using an extrapolation technique.[27] Research has been conducted to detect objects even in low light conditions such as nighttime environments. In the nighttime environment, current object proposal approaches based on a daytime circumstance show poor results, because the characteristics of vehicles, for instance, color, texture, and edges, are faded at low illumination. Kuang et al. propose an algorithm that uses the nighttime image enhancement to improve the performance of the feature extractor even in low light circumstances.[11] Detecting vehicles in images is not an easy task when various challenging lighting conditions create hindrances.

In this article, we focus on robustly detecting vehicles at harsh lightning condition, and we chose three driving environments with harsh lighting conditions. First, an urban night driving environment was selected since it includes both dynamic illumination changes and low illumination cases. Second, a suburban night driving environment was chosen that represents lighting conditions such as a low illumination case and a strong contrast case. Finally, an underpass driving environment was selected because it contains a dynamic illumination change case. These three driving environments have a large effect on vehicle detection tasks since there are difficulties resolving problems such as insufficient image information acquisition under low luminance conditions, the saturation of image information under excessive light exposure conditions, and vehicle shape modification. Thus, we employ two different image feature models and introduce a feature combination method using a soft computing technique that is appropriate for fine classification to obtain higher accuracy under challenging lighting driving conditions.

However, recent works have been devoted to studying how to handle more information effectively. Wang and Cai proposed a variant of the feature combination method using Haar and HOG features to mitigate false positive detected vehicles.[28] Another feature fusion method for three-dimensional (3-D) object retrieval based on a hypergraph was proposed by Wang et al. called the dense kernel LBP feature for describing 3-D objects.[29] Wang et al. proposed a feature fusion method to find a specific person in surveillance systems.[30] Ren et al. proposed feature fusion using the maximum margin multimodal deep neural network method to divide the image into foreground and background.[31] Kuang et al. proposed an algorithm that can detect vehicles stochastically even in low light conditions such as nighttime based on the Bayes saliency map that consists of three features.[32] In spite of the various feature fusion techniques mentioned above, delicate improvements are required in the technique for dealing with incomplete image features from driving in difficult lighting conditions.

Numerous attempts have been made to demonstrate the excellence of the preceding vehicle detection techniques. Nevertheless, the feature fusion technique seems to be lacking. It is obvious that a well-designed image feature information contributes to improving the performance of an object detection system; however, a significant difference in system performance may occur in accordance with how a method combines dissimilar image features. In some studies, a feature fusion method has been applied to vehicle detection under fine weather or illumination conditions.

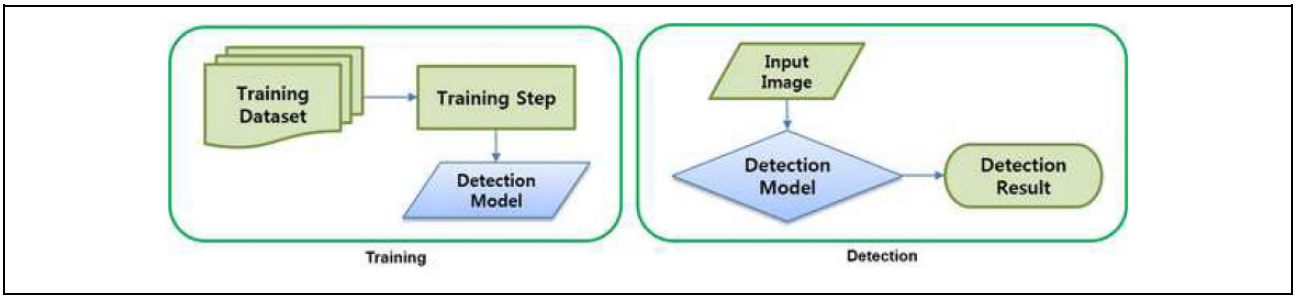

Figure 1. Conventional detection method.

Therefore, a new feature fusion method that considers the characteristics of each feature used as a source for feature fusion is required to solve the vehicle detection problem under challenging illumination conditions that cause disturbances such as unwanted noise, loss of color information, and excessive exposure to light.

In this work, we propose a vehicle detection framework based on a feature fusion method using an adaptive neuro-fuzzy inference system (ANFIS). Our method has focused on a novel feature fusion technique that combines the HOG features and the LBP features using the ANFIS to solve the degradation of the vehicle detection system under challenging environments. The proposed method achieved more accurate vehicle detection performance than that of other detection systems using standard feature fusion methods such as feature concatenation, linear combinations, and weighted fusion. The main contributions of this article are:

- a new detection framework that includes a two-phase learning structure: (1) the first phase is composed of feature models and (2) the second phase is an ANFIS model;
- the performance of the LBP feature for preceding vehicle detection under challenging lighting conditions is verified in different driving cases; and
- three driving environments under the bad lighting condition are subdivided, and the preceding vehicle detection system using the proposed framework in each subdivided driving environment shows better performance than those of methods using different types of detection processes.

The remainder of the article is organized as follows. In the "Vehicle detection framework using ANFIS" section, a vehicle detection framework that features ANFIS-based feature fusion is proposed, and the feature analysis and two-phased learning process for the frameworks are described precisely. In the "Experiments" section, the experimental results of the proposed vehicle detection system are presented and evaluated in three different challenging lighting driving environments. Finally, the "Conclusion" section concludes the article with an overall summary and suggestions for future work.

Vehicle detection framework using ANFIS

In this section, we introduce a novel vehicle detection framework based on the ANFIS, and we propose a feature combination, of HOG and LBP, method for adequate vehicle detection under challenging lighting conditions. Furthermore, we use an illuminationally transformed testing data set to evaluate the suitability of the proposed feature combination for vehicle detection in a low-intensity illumination environment. The structure of the vehicle detection framework based on the recognition model using local features is shown in Figure 1. In general, detecting desired objects in an image requires two main subsystems: a training part and a detection part. In the training part (the left side of Figure 1), a large training data set is required to build a detection model, and the output is an object model based on a local feature matching system. In the detection part (the right side of Figure 1), each image in the successive image sequence is entered into the prepared object model as an input image, and the model examines image patch windows using a sliding window method.

Overview of the proposed framework

Our detection framework is composed of two principal training parts and two built classification models as shown in Figure 2. The first is a feature model training part that yields two feature classification models; the second is a feature fusion model training part that constructs an ANFIS classification model. The upper flowchart of Figure 2 demonstrates the process of building the vehicle feature. Sufficient training data set, larger than thousands, is required to build a vehicle feature model. In our approach, we adopted two local feature descriptors, HOG and LBP. The HOG descriptor accurately represents the edge of the local regions in geometric transformations. However, the HOG descriptor has drawbacks when illumination changes. The LBP descriptor can compensate the shortage of the HOG due to LBP's robustness to changing illumination. We employ AdaBoost to train each feature model and select the decision tree model as a weak classifier of AdaBoost to classify each image sample into a binary class. Output feature models are generated after the off-line AdaBoost training step. The details of the feature information

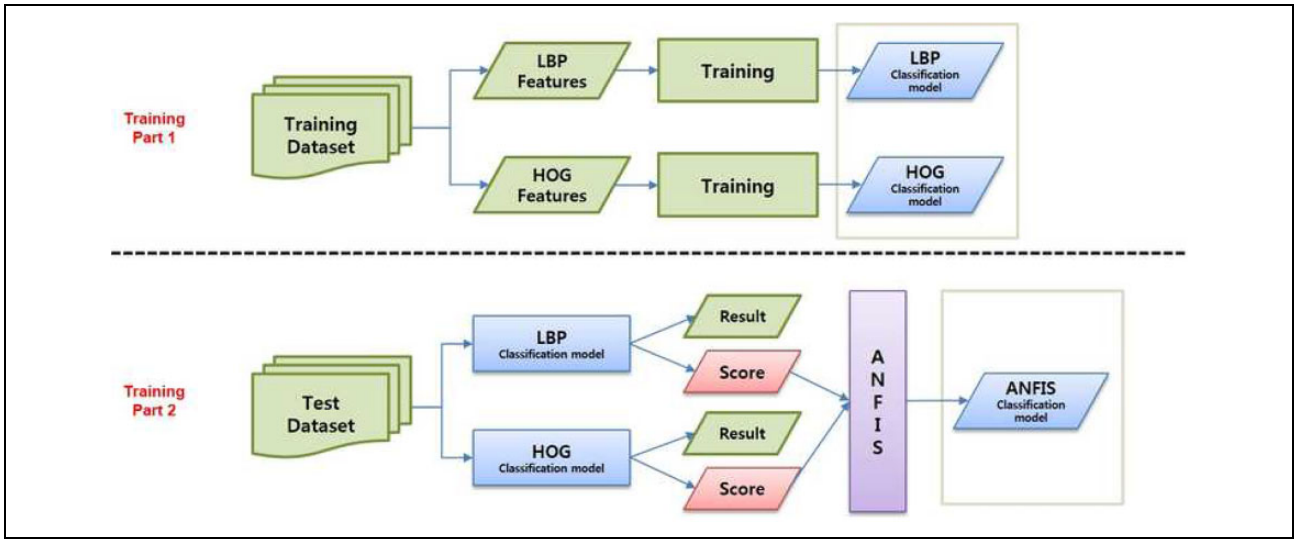

Figure 2. Flowchart of the proposed vehicle detection framework.

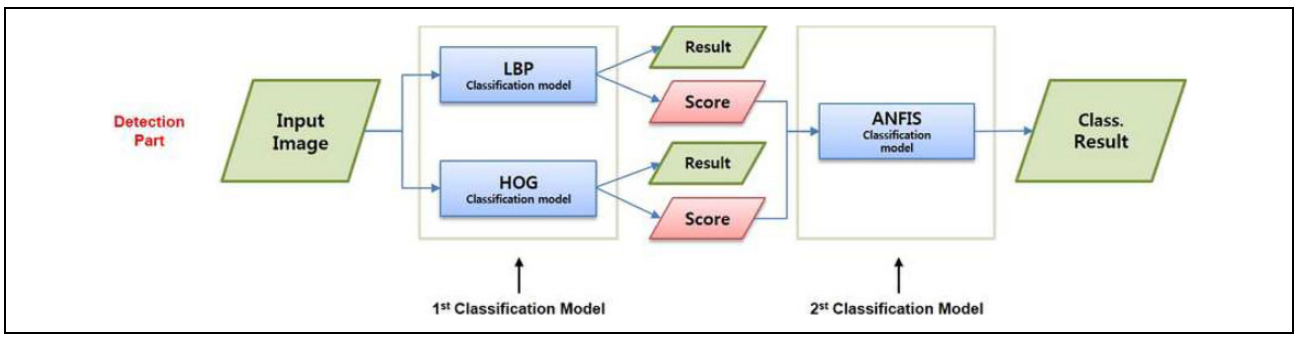

Figure 3. Detection process of the proposed method.

and the effect of the feature combination are demonstrated in the "ANFIS-based feature fusion algorithm" section.

The second part of our main framework is the feature fusion model training part. The training data set is used in both training parts to maintain consistency and connect information between the first and second parts. AdaBoost yields two outputs, a score value and the class of the input data. In contrast to conventional detection systems that use one-phase classification, such as a feature concatenation method or a linear combinations method, our feature combination approach provides fine classification results. The improvement of the algorithm has two causes: The first is the reflection of the variety of classification cases that number in the thousands and the second is the distinguishing of small differences due to the rough boundaries that come from the limitations of the feature fusion method used in them. Thus, we adopt the ANFIS model to combine two feature descriptors to solve these problems.

Figure 3 exhibits the process of the detection part; it consists of two major classification phases: one for calculating the score values of each feature model output and the other for feature fusion using the ANFIS model. The input of the detection part is an image patch window of size 24×24, and the input of the second phase classification model is the score values of each feature model. Employing the score is the main idea of our work on account of its contribution to performance improvement.

ANFIS-based feature fusion algorithm

It has been proven that the ANFIS is a successful artificial intelligent system based on the expert system, which can make inferences from data. The ANFIS, proposed by Jang,[33] features the advantages of both fuzzy logic and the neural network, constructs a clear input–output mapping structure, and provides a strong learning and decision-making process to target systems, including nonlinear systems. The principal objective when designing the ANFIS is to model a nonlinear system to predict its output. Therefore, using the ANFIS is appropriate to model manifold problems where the relationships between the internal variables of the model are unidentified. Regarding the feature fusion technique for the vehicle detection system, a principal issue to be discussed is how to determine the proportion of individual feature information used in combination. In this study, we added classification phase–based performance.

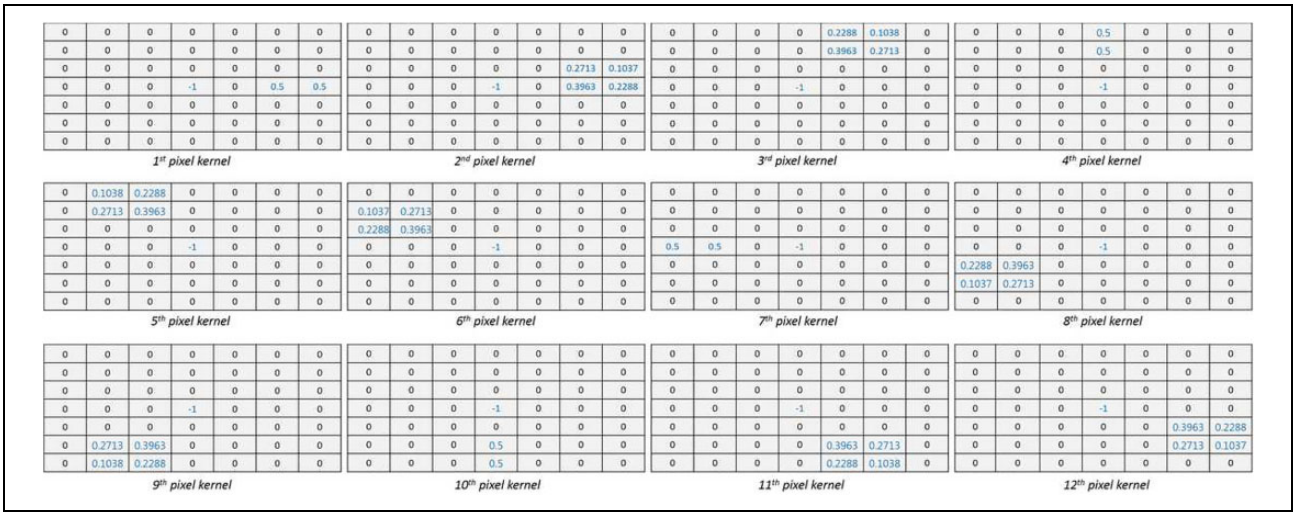

Figure 4. Filter kernels for the LBP using $P = 12$, $R = 2.5$. LBP: local binary pattern.

There are three good reasons for thinking that the ANFIS can be a good choice for a feature fusion method:

- The membership functions (MFs) of the ANFIS can act as weight functions that determine the significance of the features used in this system and reflect the vast amount of information on the correlation between the training input and the output of the sample data.
- The correlation information can be densely included as the trained MFs with slightly adjusted MF parameters, no matter how suboptimal.
- Therefore, the prediction process of the ANFIS can be adapted to classification purposes because the output prediction value can be understood as a score value for the classification job.

HOG- and LBP-based feature model

We chose two features, HOG and LBP, to build a vehicle detection model in this article. Two feature descriptors are employed to build the vehicle object model. The proposed system requires two AdaBoost classifiers for each feature model. Basically, the HOG feature extractor consists of five steps: (1) normalize the gamma and color, (2) compute the gradients, (3) weighted vote into spatial and orientation cells, (4) contrast normalization over overlapping spatial blocks, and (5) collect HOG features over the desired region. Besides, the cells are integrated into larger spatial regions to accomplish better invariance to illumination or shading by increasing the quantity of information related to the adjacent image area.

The LBP extracts the local texture information of a grayscale image and captures the appearance of an image in a small neighborhood around a center pixel. Each pixel in a cell (except the center pixel) is compared to the center pixel.

In this article, we employ variant-sized LBP cells to improve accuracy. The kernels are built using a 7×7 weight mask, each element computed by considering the weight of neighborhood pixels, as shown in Figure 4. The LBP operator function is shown in the following equation

$$V_{\mathrm{LBP}} \approx v(o(I_0 - I_{\mathrm{c}}), \cdots, o(I_p - I_{\mathrm{c}})) \tag{1}$$

where

$$o(x) = \begin{cases} 1 & \text{if} \quad x > 0 \\ 0 & \text{if} \quad x \leq 0 \end{cases}$$

Equation (1) demonstrates a vector V as the joint distribution of the intensity of $p + 1$ image pixels; p is the number of neighborhood pixels, $I_k(I_k, I_k)$ is supposed to be a pixel in a local region of a vehicle image patch, and I_{c} is the center position of the 3×3 area. The other eight pixels within the window region are $I_0, \cdots, I_p$. $I_p(p = 0, \cdots, p - 1)$ corresponds to the intensity of p equally occupied pixels on a circle radius $R(R > 0)$ that build up a circularly symmetric set of neighbors. At first, V is expressed as

$$V_{\mathrm{LBP}} = v(I_{\mathrm{c}}, I_0, \cdots, I_p) \tag{2}$$

Since $I_p - I_{\mathrm{c}}$ represents the differences, they are independent of I_{c}, and $v(I_{\mathrm{c}})$ is defined as the entire luminance of an image, equation (2) is the same as

$$V_{\mathrm{LBP}} = v(I_0 - I_{\mathrm{c}}, I_0, \cdots, I_p - I_{\mathrm{c}}) \tag{3}$$

Thus, the distribution can be represented as

$$V_{\mathrm{LBP}} = v(o(I_0 - I_{\mathrm{c}}), \cdots, o(I_p - I_{\mathrm{c}})) \tag{4}$$

The original form of AdaBoost constructs a composite classifier, called a strong classifier, by training each weak classifier. Viola and Jones[34] employ Haar features to analyze training images; however, the Haar features are

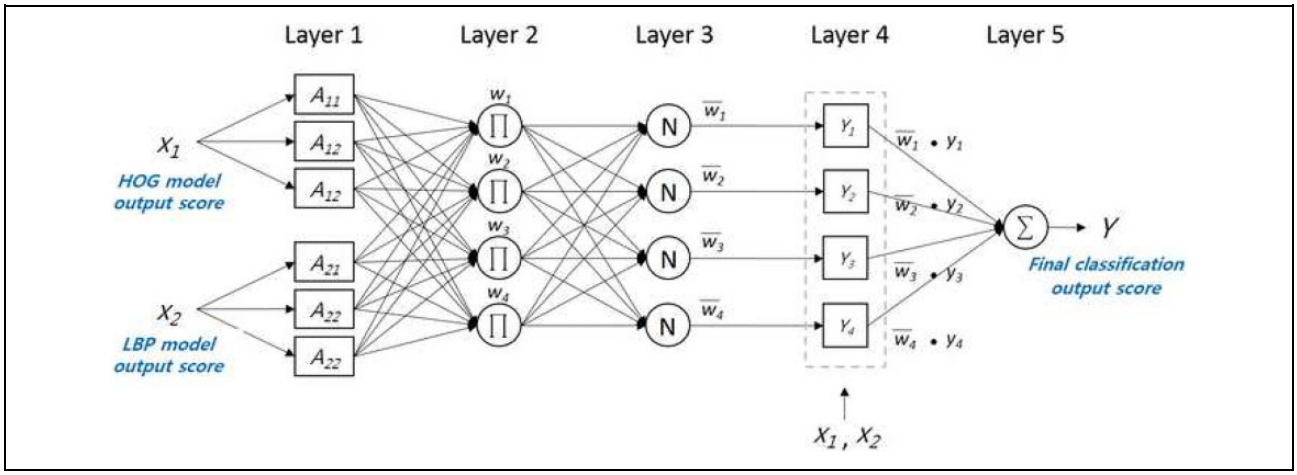

Figure 5. The ANFIS structure for feature fusion. ANFIS: adaptive neuro-fuzzy inference system.

inadequate for detecting a vehicle in a complex driving scene correctly. Thus, we consider combining the feature descriptors and AdaBoost to help the classification model adjust the weighting parameters. In the two-class classification problem, we assume that a set of n-labeled training image samples are $(x_1, y_1), \cdots, (x_n, y_n)$, where $y_i \in \{-1, +1\}$ is the class label associated with sample x_i. On each round, $t = 1, \cdots, T$, which is a distribution called Dt, is computed over m training examples, and a given weak learner or weak learning algorithm is applied to find a weak hypothesis, $h_t : \mathbb{H} \rightarrow \{-1, +1\}$, where the aim of the weak learner is to find a weak hypothesis with a low weighted error, ε_t, relative to D_t. The final or combined hypothesis, H, computes the sign of a weighted combination of weak hypotheses

$$F(x) = \sum_{t=1}^{T} \alpha_t h_t(x) \tag{5}$$

This is equivalent to saying that H is computed as a weighted majority vote of the weak hypotheses, h_t, where each is assigned weight α_t. For the front vehicle detection, x_i is an image sub-window of fixed size (24 × 24 in our system) containing the vehicle $(y_i = +1)$ or non-vehicle $(y_i = -1)$. Finally, the n weak classifiers constitute one stronger classifier in the form of a linear combination. In AdaBoost, each example of x_i is associated with a weight of w_i, and the weights are updated iteratively using a multiplicative rule according to the errors in the previous learning, so that greater weight is given to weak learners with lower errors.

Feature fusion based on ANFIS

In this section, we are concerned with using the ANFIS as a feature fusion method. An ANFIS-based feature fusion technique is an excellent method due to its artificial intelligent characteristics. In the second classification phase of the detection process, the ANFIS determines how to fuse the weight of the scores of the feature model output. There are two score outputs transferred from the first classification phase by pushing input window images through HOG and LBP. In a simple two-class classification system, a score value is sorted as one of the two classes by judging whether the output is more or less than a threshold value, where it is normally a center value between two class label values. However, if an additional classification operation has to be combined with an existing one, it requires a feature fusion method that determines which feature model is more trustworthy. Thus, the ANFIS performs a significant role in determining which feature's classification result is more trustworthy. This task is crucial for a feature fusion method, since it highly affects the system performance based on the fusion method. The structure and the detail parameters of the proposed ANFIS model are represented in Figure 5.

After processing the ANFIS and the entire structure of the two-phase classification, we may now turn to the main issue, which is the necessity of the second classification phase. This is the most important advantage of using second phase classification for feature fusion. Several studies have proposed a combination of multiple features to compensate for the weakness of other models[11,31,35]; however, how to cover several special cases for which the classification model provides a misclassified output is an open problem. Thus, we must consider methods to reduce the classification error caused by the occasion mentioned above.

In Figure 6, the specific occasion is described in detail. Figure 6 exhibits the big differences between the conventional and proposed feature fusion methods. The first is that the conventional method has no second phase classification part, and the second is that the fusion step of the conventional method is placed in advance of the first classification phase, while that of the proposed method is behind the first classification phase. Moreover, our method combines the score values that are the output information of each feature

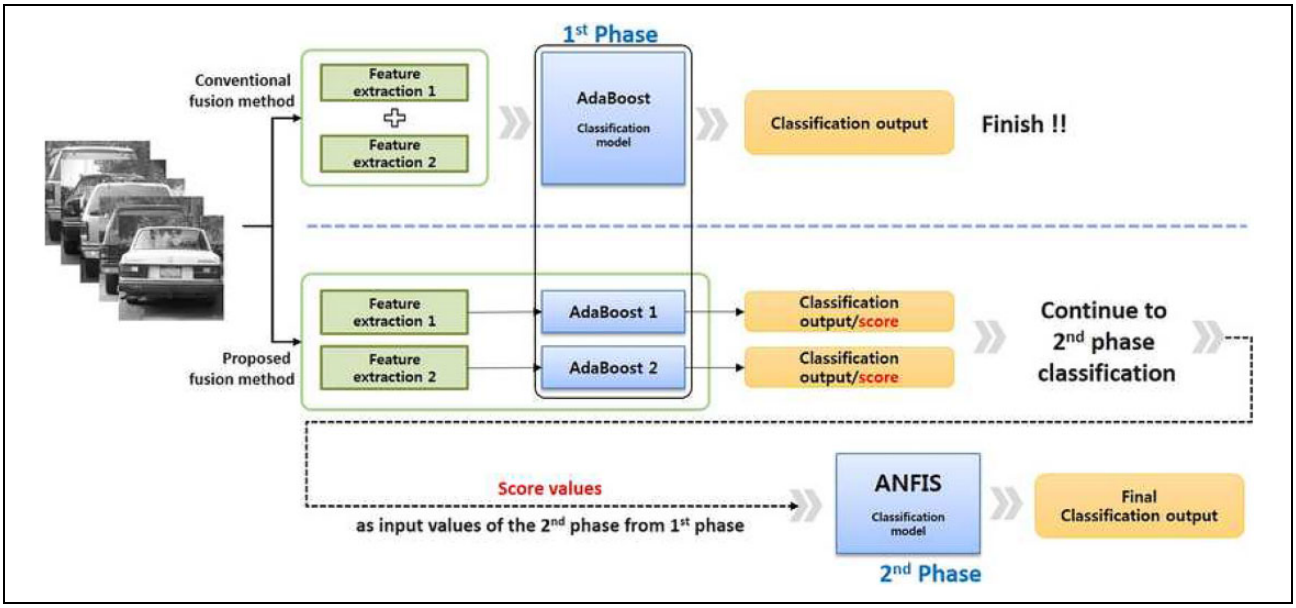

Figure 6. The comparison of the structures of the conventional feature fusion method and the proposed feature fusion method.

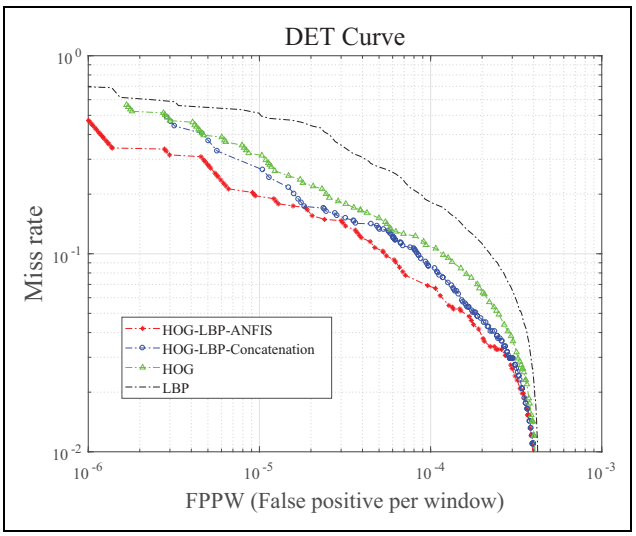

Figure 7. Vehicle detection accuracy comparison (general performance).

model, where each takes a different approach from each other. In the classification step, there can be misclassifications due to the incompleteness of the classification model. If an object detection system only uses one classification phase, it naturally generates classification errors due to the input data that are located close to the classification boundary, even though the system employs a classification model based on a new feature that fuses multiple features. Therefore, it is difficult for the one-phase classification method to prevent misclassification, and there is no chance to correct errors after the classification phase. In contrast, our two-phase classification method has one more learning step based on the ANFIS. Using the score values as inputs in the second phase training step that have the same output labels

Table 1. Parameters of HOG and LBP. HOG: histogram of oriented gradients; LBP: local binary pattern.

HOG		LBP	
Window size	24 × 24	Mask size	9 × 9
Cell size	8 × 8	Number of neighbor pixels	12
Number of cells in the block	2 × 2	Circle radius	2.5
Number of bins	9		

HOG: histogram of oriented gradients; LBP: local binary pattern.

allows our two-phase classification method to correct the second phase step. This technique has a strong advantage when all two-features-or-more combined feature models in the first classification phase result in misclassifications. A first-phase-only system[11,31,35] cannot compensate for classification errors later; however, the second phase classification model in our approach can correct the wrongly classified results of the previous phase. The point we wish to emphasize is the difference between the conventional method and ours, which is whether they have the ability to correct misclassified decisions to improve system performance.

Experiments

This section reports the experiments conducted to identify the effectiveness of our vehicle detection framework based on the proposed feature fusion method. Errors were calculated using the definition of the detection error trade-off (DET) curve, which is based on the classification performance of an object detection system; this will be explicated in the "Experimental setting" subsection. We assumed

Figure 8. Driving environment for case 1.

several lighting conditions that disturb smooth vehicle detection to examine challenging illumination cases, such as a dynamic illumination change case, a low illumination case, a strong contrast case, and an abrupt illumination change case. Here, we conducted three vehicle detection experiments in different challenging driving environments. First, an urban night driving environment was selected, since it includes both dynamic illumination changes and low illumination cases. Second, a suburban night driving environment was chosen, which represents lighting conditions such as low illumination cases and strong contrast cases. Finally, an underpass driving environment was included because it contains dynamic illumination change cases.

Experimental setting

DET curves are used to evaluate the performance of our vehicle detection framework. DET curves represent miss rates on the y-axis against false positives per window (FPPW) on the x-axis using a logarithmic scale. The miss rate is defined as

$$\text{Miss rate} = \frac{\#\ \text{False Negatives}}{\#\ \text{True Positives} + \#\ \text{False Negatives}} \quad (6)$$

When evaluating these plots, it is important to keep in mind that they characterize the behavior of the classifier, since the classifiers act as a basic part of the whole detection system. The DET curves located closer to the bottom left corner are considered as showing better performance and miss rate value with respect to 10^{-4} FPPW on the x-axis, and are widely used as a reference performance point in the field when comparing other detectors. The first DET curve plot is shown in Figure 7.

We compared the miss rate of our vehicle detection framework with that of the three different systems based on the feature model using the HOG–LBP with conventional feature fusion (HLCFF) method, HOG, and LBP, respectively, to test the proposed vehicle tracking framework. The conventional feature fusion method used in HLCFF is feature concatenation, which is a basic and widely used approach in this area. We investigate the

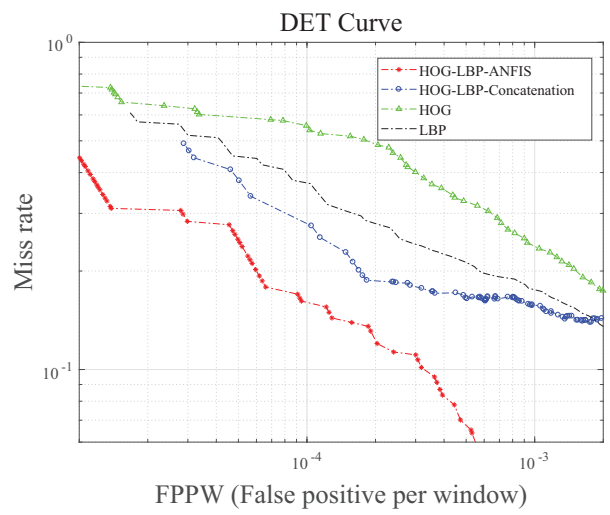

Figure 9. Vehicle detection accuracy comparison under case 1 environment.

performance differences between the detection methods by the DET curve for every obtained test image.

We evaluate the performance differences between the proposed framework and different algorithms without the ANFIS-based feature fusion technique for various challenging lighting scenarios in three driving environments. Unfortunately, there are few adequate and publicly opened available test image data sets that properly show the characteristics of our vehicle detection framework in the proposed scenarios, since open data samples are mostly recorded in daylight, which does not reflect harsh illumination driving conditions. Thus, we recorded videos containing the preceding vehicle images under challenging light environments using a black box camera installed behind the rearview mirror in the car. We filmed the videos that were recorded in three different road environments to diversely express the characteristics of the types of illumination changes affecting vehicle detection when driving: (1) an urban night driving environment, (2) a suburban night driving environment, and (3) an underpass driving environment. Test images from the videos filmed in the three

Figure 10. Driving environment for case 2.

scenarios are used for experiments in the following sections to compare the proposed method with others.

Experimental results

Before we show the experimental results under challenging illumination cases, an experiment under normal circumstances is conducted to identify the general performance of the proposed framework. We set the parameters of HOG features and LBP features that are shown in Table 1. We added the linear interpolation and normalization process to the LBP feature computation to increase the accuracy of the LBP. It is noted that we employed vehicle image data sets,[36,37] comprising 3425 positive images of vehicle rears taken from different perspectives and 3900 negative images that do not contain vehicles. These are extracted from road sequences with normal illumination conditions, which we consider as representing normal driving environments.

Experimental result 1: Urban night road scene. Figure 8 depicts the harsh lighting condition of an urban night environment, where the driving scene on the left side in Figure 8 is captured from the footage that we filmed when driving in the city at night. The preceding vehicle images on the right side in Figure 8 are used as the positive sample set for training and testing the system, which is extracted from captured images similar to the left one in Figure 8.

The comparison results in the driving environment shown in Figure 8 are plotted in Figure 9. The red line indicates the DET curve of our algorithm based on the ANFIS feature fusion method and the blue, black, and green lines indicate the detection performance with HLCFF, HOG, and LBP, respectively. As shown in Figure 9, our detection method outperforms the others in the first challenge.

Experimental result 2: Suburban night road scene. Figure 10 shows a driving scene on a suburban night road containing harsh lighting conditions, where the sample image windows include vehicle shape, and Figure 11 presents DET curves for the four detection methods. In Figure 10, it can be verified that the contrast in the vehicle region is very strong due to the headlights of the following vehicle and

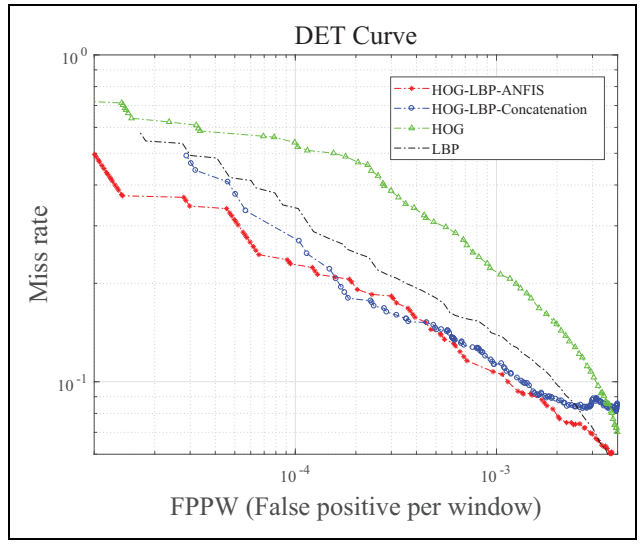

Figure 11. Vehicle detection accuracy comparison under case 2 environment.

that, in a rough manner, this presents the bulk of the two antithetical illumination cases: very bright and very dark. In this scenario, our method shows the best performance at 10^{-4} FPPW and at most other FPPW steps. According to the result, feature fusion with image features such as LBP, which is robust to illumination changes, has a positive effect on vehicle detection under difficult lighting conditions, as shown in Figure 11.

Experimental result 3: Underpass road scene. In the third experiment, a preceding vehicle is entering or leaving an underpass in daylight, as shown in Figure 12. Figure 13 verifies that our detection framework performs the best among the four detection methods at 10^{-4} FPPW, which is widely considered a reference point. In contrast to the previous two challenging lighting cases, vehicle detection in an underpass road scene intensively focuses on the strong contrast and abrupt contrast changes in the image. From the perspective of the following vehicle detection algorithm, dynamic contrast changes in the vehicle region in the image windows are a critical disturbance factor. It is a heavy task for the system to make a correct classification

Figure 12. Driving environment for case 3.

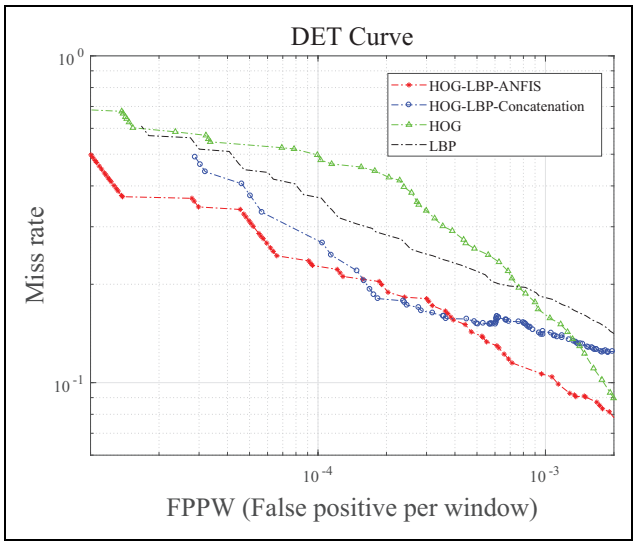

Figure 13. Vehicle detection accuracy comparison under case 3 environment.

decision when the illumination intensity near a target vehicle region rapidly increases or decreases. In an extremely bad circumstance as in the third scenario (without sufficient illumination), the proposed vehicle detection framework can still detect a vehicle's presence in the image window correctly.

Conclusion

This article proposed a novel object detection framework using an ANFIS-based feature fusion technique for use in vehicle detection under challenging lighting environments. Our approach models two vehicle shape articulations based on feature descriptors, HOG and LBP, and one feature fusion process of combining the output scores of each model. The entire detection framework is divided into two phases, the first of which includes two vehicle feature models and the second of which includes the ANFIS-based feature fusion model. In comparison to previous conventional vehicle detection frameworks that construct a classification model based on a single feature model or a concatenated feature fusion model, our approach achieves superior performance in three different severe illumination conditions. In particular, when a vehicle detection test is conducted in an urban night driving environment, our method showed outstanding performance, with miss rates under 2×10^{-1} at the reference FPPW point. In addition, the ANFIS-based feature fusion concept in our detection framework can extend a vehicle detection framework to more variations using feature fusion methods based on different soft computing methods or combinations of other newly developed image features.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This paper is supported by basic science research program through the National Research Foundation of Korea funded by the Ministry of Education under Grant (NRF-2016R1D1A1B01016071) and also (NRF-2016R1D1A1B03936281).

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